Informatica® Cloud Application Integration
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Amazon Bedrock Chat with Image File using Guide

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Preface

Use *Amazon Bedrock Chat with Image File using Guide* to learn how to ask questions to the Amazon Bedrock Large Language Model (LLM) based on an uploaded image file's context using a guide. This guide assumes that you have an understanding of the Amazon Bedrock Connector concepts.

CHAPTER 1

Introduction to Amazon Bedrock Chat with Image File using Guide recipe

The Amazon Bedrock Chat with Image File using Guide recipe is based on REST and SOAP APIs. You can ask questions to the Amazon Bedrock Large Language Model (LLM) based on an uploaded image file's context.

The recipe allows you to upload an image file with context, provide a user prompt, read the image context, and answer the user's questions based on the context of the image file using the Amazon Bedrock Large Language Model (LLM).

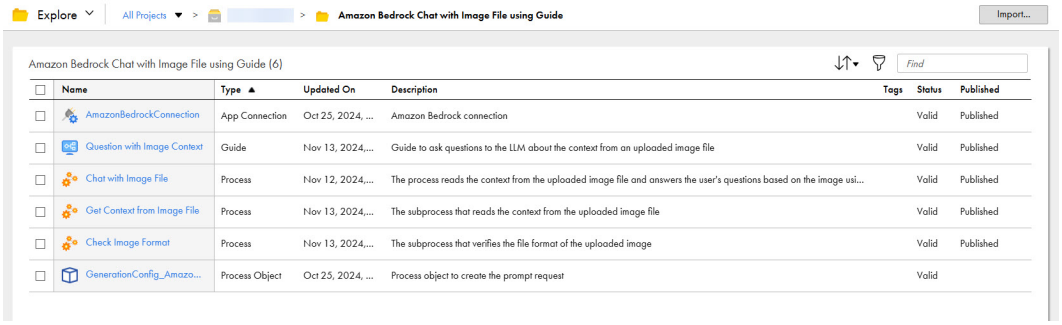
You can upload files of the following formats:

`.jpeg`, `.png`, `.gif`, or `.webP`.

Amazon Bedrock Chat with Image File using Guide recipe contents

The Amazon Bedrock Chat with Image File using Guide recipe contains a process object, an app connection, a guide, and three processes.

The following image shows the assets that the Amazon Bedrock Chat with Image File using Guide recipe package contains:



Amazon Bedrock Chat with Image File using Guide recipe assets

The following table lists the assets that the Amazon Bedrock Chat with Image File using Guide recipe package contains:

Asset Name	Asset Type	Description
GenerationConfig_AmazonBedrock_PO	Process object	Process object that creates the prompt request.
AmazonBedrockConnection	App connection	Amazon Bedrock connection.
Get Context from Image File	Process	Subprocess that reads the context from the uploaded image file.
Check Image Format	Process	Subprocess that verifies the format of the uploaded image file. The supported file formats are `.jpeg`, `.png`, `.gif`, and `.webP`.
Chat with Image File	Process	Process that reads the context from the uploaded image file and answers the user's questions based on the image using the Amazon Bedrock Large Language Model (LLM).
Question with Image Context	Guide	Guide that allows users to ask questions to the LLM about the context from an uploaded image file.

CHAPTER 2

Using the Amazon Bedrock Chat with Image File using Guide recipe

To use the Amazon Bedrock Chat with Image File using Guide recipe, you must perform the following steps manually:

1. Copy and access the recipe.
2. Configure and publish the Amazon Bedrock connection.
3. Configure and publish the processes.
4. Publish and run the guide.

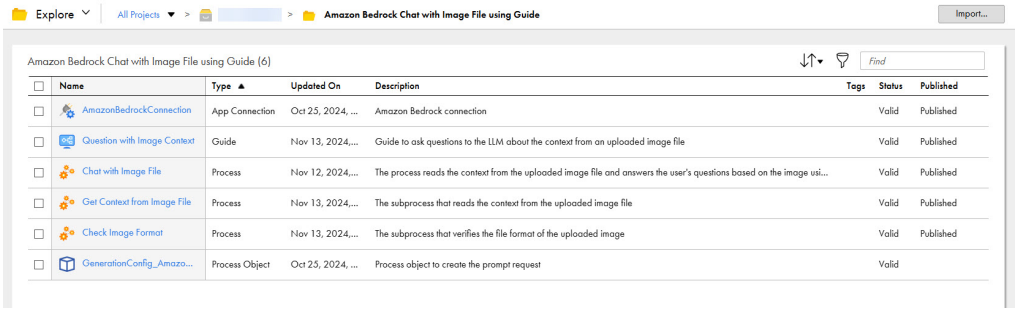
Step 1. Copy and access the recipe

Copy the pre-configured assets in the recipe to a separate project or folder.

1. Open the **Amazon Bedrock Chat with Image File using Guide** recipe and click **Use**.
2. Select the location where you want to copy the recipe, and then click **Continue**.
3. In the **Copying the recipe** dialog box, click **OK**.

   It might take some time for the recipe to get copied. You will receive a notification when the recipe is ready for use.
4. After the recipe is copied, click **Explore** to access the recipe content.
5. Navigate to the project or folder where you copied the recipe or enter the recipe name in the **Find** box.

   All the assets in the recipe are displayed as shown in the following image:



Step 2. Configure and publish the Amazon Bedrock connection

Configure the access key, secret key, and region to connect to Amazon Bedrock, and then publish the AmazonBedrockConnection connection.

1. Open the **AmazonBedrockConnection** connection.
2. In the **Connection Properties** section, enter values for the following properties:

Property	Description
AccessKey	The access key to access Amazon Bedrock.
SecretKey	The secret key to access Amazon Bedrock.
Region	The region that hosts Amazon Bedrock.

3. Save and publish the connection.

Step 3. Configure and publish the processes

Configure the deployment details of the LLM model and publish the processes.

1. To publish the **Get Context from Image File** process, click **Actions** in the row that contains the process and select **Publish**.
2. To publish the **Check Image Format** process, click **Actions** in the row that contains the process and select **Publish**.
3. Open the **Chat with Image File** process.
4. On the **Temp Fields** tab of the Start step, in the **Model_id** field, enter the deployment ID of the LLM model that was deployed.
5. Optionally, in the **Configure Request Parameters** step, enter the prompt instructions in the **Assignments** field by updating the **Prompt_Configuration** field using the Expression Editor, as shown in the following sample code:

```
For Prompt_Configuration:
<GenerationConfig_AwsBedrock_PO-1>
   <Temperature>0.5</Temperature>
   <TopP>0.6</TopP>
   <MaxTokens>500</MaxTokens>
</GenerationConfig_AwsBedrock_PO-1>
```

For the **Prompt_Configuration** field, enter values for the following properties:

Property	Description
temperature	Controls the randomness of the model's output. A lower value close to 0 makes the output more deterministic, while a higher value close to 1 increases randomness and creativity. For example, if `temperature` is set to 0.5, the model balances between deterministic and creative outputs.
topP	Determines the cumulative probability threshold for token selection. The model considers the smallest set of tokens whose cumulative probability meets or exceeds `topP`. For example, if `topP` is set to 0.1, the model considers only the top 10% most probable tokens at each step.
max_tokens	Defines the token count of your prompt. The value can't exceed the model's context length. Most of the models have a context length of 2048 tokens.

6. Optionally, in the **Prepare Request** step, enter the prompt instructions in the **Assignments** field by updating the **Request** field using the Expression Editor, as shown in the following sample code:

```
For Request:
<root xmlns:m="urn:informatica:ae:xquery:json2xml:meta-data">
<messages m:isArray="true">
    <role>user</role>
    <content m:isArray="true">
        <text>Image: </text>
    </content>
    <content m:isArray="true">
        <image>
            <format>jpeg</format>
            <source>
                <bytes>{$temp.Context_From_File }</bytes>
            </source>
        </image>
    </content>
        <content m:isArray="true">
        <image>
            <format>png</format>
            <source>
                <bytes>{$temp.Context_From_File }</bytes>
            </source>
        </image>
    </content>
    <content m:isArray="true">
        <image>
            <format>gif</format>
            <source>
                <bytes>{$temp.Context_From_File }</bytes>
            </source>
        </image>
    </content>
    <content m:isArray="true">
        <image>
            <format>webp</format>
            <source>
                <bytes>{$temp.Context_From_File }</bytes>
            </source>
        </image>
    </content>
    <content>
        <text>{$input.User_Prompt }</text>
    </content>
</messages>
<interfenceConfig>
<max_tokens m:type="xs:int">{$temp.Prompt_Configuration[1]/MaxTokens }</max_tokens>
<temperature m:type="xs:double">{$temp.Prompt_Configuration[1]/Temperature }</
temperature>
```

```
<topP m:type="xs:double">{$temp.Prompt_Configuration[1]/TopP }</topP>
</interfenceConfig>
</root>
```

7. Save and publish the process.

Step 4. Publish and run the guide

You can ask questions to the LLM about the context from an uploaded image file. You can use files of the following formats:

`.jpeg`, `.png`, `.gif`, or `.webP`.

To publish and run the guide, perform the following steps:

1. Open the **Question with Image Context** guide.
2. On the **Start** tab of the Start step, ensure that the **Run As** field is set to **Current User**.
3. Save and publish the guide.
4. On the **Actions** menu, click **Run**. Alternatively, you can copy the execution URL from the **Properties Details** dialog box to run the guide.
5. On the **Instructions** page, select the image file that you want to upload from your machine.
6. In the **Enter your query below** field, enter a question based on the file context.
7. Click **Continue**.
   You will receive a response from the LLM.
8. To ask the next question, click **Next Question**.
9. Enter new questions with the same context and click **Continue**.

You can also use the embed code to embed the guide into an HTML document of a third-party application.